Exp 12b – Determination of Citric Acid in Soda

INTRODUCTION

A titration is a way of carefully controlling the amounts of reactants added in a chemical reaction, so that the amount or concentration of one of the reactants may be determined from the amount and/or concentration of the other. In this experiment, the concentration of citric acid in lemon-lime soda will be determined by titration (reaction) with sodium hydroxide.

EXPERIMENTAL BACKGROUND

In a titration, one reactant is placed in a buret (titrant), while the other is placed in an Erlenmeyer flask. In this experiment, sodium hydroxide will be the titrant and the acid that it will react with will be placed in the Erlenmeyer flask.

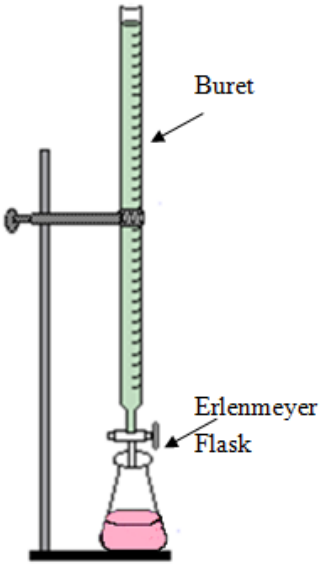


Figure 1:
Titration Apparatus

A buret has a measurement scale that starts with the zero mark near the top and then values increase downward. When reading a buret, you are recording the location of the meniscus. Later, the amount of titrant used will be calculated by taking the difference (final – initial) of the buret readings. When reading a buret, it is important to be eye-level with the buret. When at eye-level, the lines on the buret will be parallel. It is useful to hold a white piece of paper behind the buret to make the measurement scale clear. A thick black bar on the paper helps to highlight the bottom of the meniscus. In Figure 2, notice how "39" is at the top of the photograph, and "41" is at the bottom. The bottom of the meniscus is just below the "40". The smaller calibration lines represent 0.1 mL increments. The line below "40" represents 40.1 mL. The bottom of the meniscus is slightly less than halfway between "40.0" and "40.1" and therefore will be recorded as 40.04 mL.

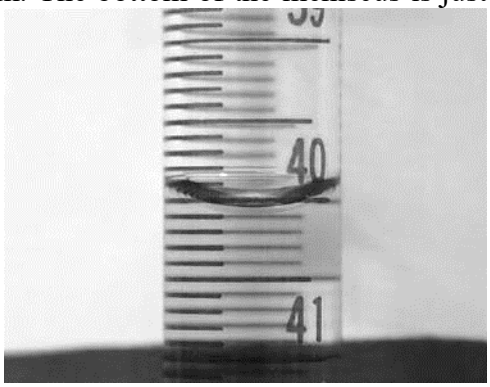


Figure 2:
Close-up of buret

Part I- Standardization

The first part of many titration experiments is the determination of the concentration of the titrant, called *standardization*. This is especially necessary for sodium hydroxide solutions, as they will react with carbon dioxide in the air, which lowers the molarity over time.

In this experiment, the molarity of a sodium hydroxide solution will be determined by reaction with oxalic acid as shown below.

$$H_2C_2O_4 \cdot 2H_2O(aq) + 2NaOH(aq) \rightarrow Na_2C_2O_4(aq) + 4H_2O(l) \quad (15)$$

Part II- Determination

In the second part of the experiment, called determination, we will use the concentration of the sodium hydroxide from part I to determine the concentration of citric acid in a lemon-lime soda (6-Down Soda). The reaction of sodium hydroxide with citric acid is shown below.

$$H_3C_6H_5O_7(aq) + 3NaOH(aq) \rightarrow Na_3C_6H_5O_7(aq) + 3H_2O(l) \quad (16)$$

If a solid is used in a titration, then water is added to the flask to dissolve the solid. Water is not a reactant. An indicator is added to the flask that will change color when the acid has completely reacted (endpoint). Proper operation of the buret will allow the experimenter to add just the right amount of sodium hydroxide to react with the acid.

The indicator used in this experiment is called phenolphthalein. Phenolphthalein is colorless in acidic conditions but turns pink in basic conditions. This means that the slightest excess of sodium hydroxide will turn the color of the solution into a light pink (endpoint). (Any additional sodium hydroxide will turn the solution darker pink and represents a significant amount of experimental error.)

ADDITIONAL READING

- Solution Stoichiometry
- Acid-Base Titrations

SAFETY EQUIPMENT REQUIRED FOR LAB

- Goggles
- Lab coat
- Nitrile gloves (provided)

SUPPLY LIST

Part I and II:

- 400 or 600-mL beaker (temporary waste container)
- 25-mL buret
- Bench stand
- Buret clamp
- 125-mL Erlenmeyer flask(s)
- Wash bottle filled with DI water

Part I:

- 10-mL graduated cylinder
- Spatula

Part II:

- 100-mL beaker
- 10-mL volumetric pipet
- Pipet bulb or pump

CLASSROOM SUPPLY AND REACTANT LIST

- NaOH (~0.1 M)
- Phenolphthalein (1 %)

Part I:

- $H_2C_2O_4 \cdot 2H_2O$ (solid)

Part II:

- Lemon-lime soda (6 Down)

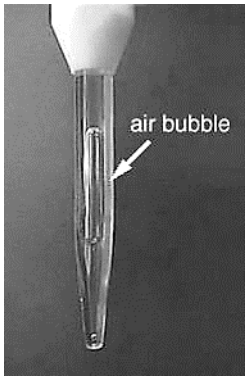


Figure 3:
Hold this black bar behind the buret, below the meniscus to make it easier to read.

PROCEDURE

Setup and Cleaning the Buret

1. ☐ Set-up the titration apparatus as shown in Figure 1.
2. ☐ Obtain a large beaker to use at your desk for temporary waste collection.
3. ☐ Rinse your buret three times using a wash bottle filled with DI-water (5 mL each) and then three times with the sodium hydroxide (NaOH) solution (5 mL each) and drain into the waste beaker before filling your buret with the NaOH solution. This ensures that only the sodium hydroxide solution is in your buret.
4. ☐ Place your buret in the clamp. Place the large beaker under the tip of the buret. With the stopcock of the buret in the open position, add several mL of NaOH. When the NaOH begins to flow out of the tip, close the stopcock. This should force out any air bubble in the tip and fill the tip with NaOH.
   a. If there is an air bubble, you may remove it by fully opening the stopcock and allowing the bubble to pass through the opening. (If this doesn't work, consult your instructor for help.)



Part I- Standardization

5. ☐ Fill your buret with ~0.1 M NaOH to between the 0-3mL marks near the top of the buret, but below the 0.00 mL mark.
6. ☐ Record your initial buret reading, being sure to estimate one decimal place beyond the smallest division.
7. ☐ Bring your Erlenmeyer flask with you to the balance.
   a. Place weigh paper on the balance and tare (zero) the balance. The mass should read 0.0000 g.
   b. Measure out between 0.1000 and 0.1200 grams of oxalic acid dihydrate onto the weighing paper and record the precise mass on your report page.
   c. Pour the oxalic acid into the flask, being careful not to spill.
8. ☐ Use a graduated cylinder to transfer approximately 10 mL of deionized water to your clean, well-rinsed Erlenmeyer flask, and swirl to dissolve the oxalic acid.

   * If it does not completely dissolve at this point, it is not a huge problem. Just be sure that it dissolves shortly into the titration, well before the endpoint.
9. ☐ Add 2 to 3 drops of phenolphthalein indicator to the flask.

10. ☐ Place a white piece of paper under your Erlenmeyer flask to better see the color change of the endpoint.
11. ☐ While swirling the flask, quickly add 10 mL of NaOH from the buret.
12. ☐ After 10-mL of NaOH, add the NaOH more slowly.
    a. Watch the center of the flask as the NaOH is added. You should be able to see a pink color where the NaOH drops land. Towards the endpoint it will take more time for the pink color to dissipate back to colorless. The endpoint is reached when a persistent faint-pink color is obtained throughout the solution.
    b. Towards the end add fractions of a drop by quickly rotating the stopcock 180° until a lasting faint pink color has been reached. Alternatively, you can get a partial drop hanging off the edge of the tip of the buret, and then touch the drop to the inside of the flask to knock it in.
13. ☐ Read and record your final buret volume.
14. ☐ Repeat steps 5-13 for the number of trials indicated by your instructor. *Redo titrations that are clearly overdone.* (Turn bright pink at the end).

    **Do not drain your buret once you are done with Part I. Refill with NaOH for Part II. **

<u>Part II- Determination</u>

15. ☐ Obtain about 40 mL of soda in a small beaker. (The soda in your lab has been opened days in advance to assure that it is de-carbonated.)
16. ☐ Use a 10-mL volumetric pipet to transfer 10.00 mL of soda to your clean, well-rinsed Erlenmeyer flask. Record the precise volume of soda used (10.00 mL).
17. ☐ Add 2 to 3 drops of phenolphthalein indicator to the flask.
18. ☐ Fill your buret with ~0.1 M NaOH to between the 0-3mL marks near the top of the buret. Record your initial buret reading being sure to estimate one decimal place beyond the smallest division.
19. ☐ Titrate the soda as outlined above in steps 10-12.
20. ☐ Read and record your final buret volume.
21. ☐ Repeat this process for the number of trials indicated by your instructor. *Redo titrations that are clearly overdone.* (Turn bright pink at the end).

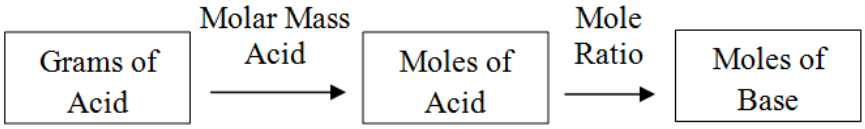
CLEANUP AND WASTE DISPOSAL

1. ☐ Pour your waste beaker into the gigantic aqueous inorganic waste beaker located in the fume hood. Your instructor will ensure that the waste is neutral before it is poured down the drain.

CALCULATIONS

Part I: Standardization

1. ☐ Use the now known grams of oxalic acid to calculate the moles of the sodium hydroxide used.



$$H_2C_2O_4 \cdot 2H_2O(aq) + 2NaOH(aq) \rightarrow Na_2C_2O_4(aq) + 4H_2O(l) \qquad (17)$$

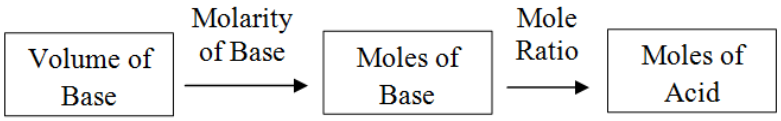
2. ☐ Calculate the net mL of NaOH added. Convert to L.
3. ☐ Calculate the molarity of sodium hydroxide from the moles of sodium hydroxide and the L of sodium hydroxide used.

$$\frac{Moles\ Base\ (from\ above\ conversion)}{Liters\ Base\ (from\ experiment)} = Molarity\ of\ Base$$

$$H_2C_2O_4 \cdot 2H_2O(aq) + 2NaOH(aq) \rightarrow Na_2C_2O_4(aq) + 4H_2O(l) \qquad (0.1)$$

Part II: Determination

1. ☐ Calculate the net mL of NaOH added. Convert to L.
2. ☐ Use the net L of NaOH, and average molarity of NaOH determined from Part I to calculate the mol of citric acid reacted.



$$H_3C_6H_5O_7(aq) + 3NaOH(aq) \rightarrow Na_3C_6H_5O_7(aq) + 3H_2O(l) \qquad (24)$$

3. ☐ Calculate the molarity of citric acid from the moles of citric acid and the volume of soda.

$$H_3C_6H_5O_7(aq) + 3NaOH(aq) \rightarrow Na_3C_6H_5O_7(aq) + 3H_2O(l) \qquad (0.1.1)$$

$$\frac{Moles\ Acid\ (from\ above\ conversion)}{Liters\ Acid\ (from\ experiment)} = Molarity\ of\ Acid$$

•

1. ☐ Use the volume of the soda can in liters, and molarity of citric acid in 7-up ® to calculate the moles of citric acid in one can of soda. Convert moles to grams.

DATA AND REPORT – DUE ____ / ____ / ____

SECTION _____

* Show your work and keep track of significant figures.

DATE _____

PARTNER _____

REFERENCE VALUES (FROM PRE-LAB QUIZ)

Oxalic acid dihydrate formula _____

Oxalic acid dihydrate molar mass _____

PART I- STANDARDIZATION

	Trial 1	**Trial 2**	**Trial 3**	**Trial 4 (optional)**
Mass of oxalic acid				
Initial Buret Reading (NaOH)				
Final Buret Reading (NaOH)				

* Show all work for trial 1on the next page.

	Trial 1	**Trial 2**	**Trial 3**	**Trial 4**
Moles of sodium hydroxide				
Net Volume NaOH used (L)				
Molarity of sodium hydroxide				
Average Molarity				

Calculations Part I: Standardization:

Show your mathematical setup for ***Trial 1*** for each of the following, complete with units:

Moles of sodium hydroxide:

Net Volume NaOH used, in liters:

Molarity of sodium hydroxide:

PART II- DETERMINATION

NaOH Concentration _____ (from standardization procedure)

	Trial 1	**Trial 2**	**Trial 3**	**Trial 4 (optional)**
Volume of "6 Down" Soda				
Initial Buret Reading (NaOH)				
Final Buret Reading (NaOH)				

* Show all work for trial 1on the next page.

	Trial 1	**Trial 2**	**Trial 3**	**Trial 4**
Volume NaOH used (L)				
Moles of Citric Acid in "6 Down" Soda				
Molarity of Citric Acid in "6 Down" Soda				
Average Molarity in "6 Down" Soda				

Additional Calculations

"6 Down" is about 10 times more concentrated in citric acid than 7 Up®, so to calculate the realistic citric acid concentration in 7 Up®, just divide by 10.

Average Molarity in 7 Up® Soda	
Grams Citric Acid in one can of 7 Up® (355 mL)	

Calculations for Part II: Determination

Show your mathematical setup for ***Trial 1*** for each of the following, complete with units:

Net Volume NaOH used, in liters:

Moles of citric acid:

Molarity of citric acid in 6-Down Soda:

Molarity of citric acid in 7-Up*:

** "6 Down" is about 10 times more concentrated in citric acid than 7 Up®, so to calculate the realistic citric acid concentration in 7 Up®, just divide by 10.*

Grams Citric Acid in one can of 7 Up® (355 mL):